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Impact of preprocessing in data integration of single-cell RNA-seq data

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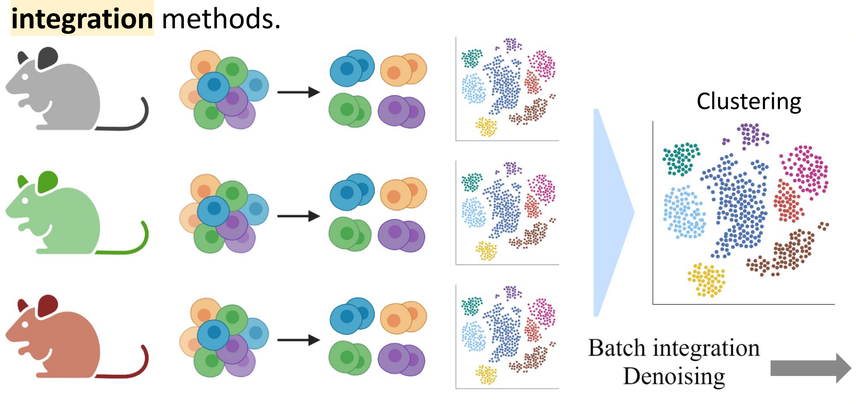
Introduction

Over the last years, numerous single-cell data analysis tools have been introduced as **de-noising** or **batch integration** methods.



Here, we found that most single-cell studies employed various preprocessing steps without reasoning. This fact is particularly alarming since these read count transformations can significantly alter data distribution and affect downstream cell clustering results.

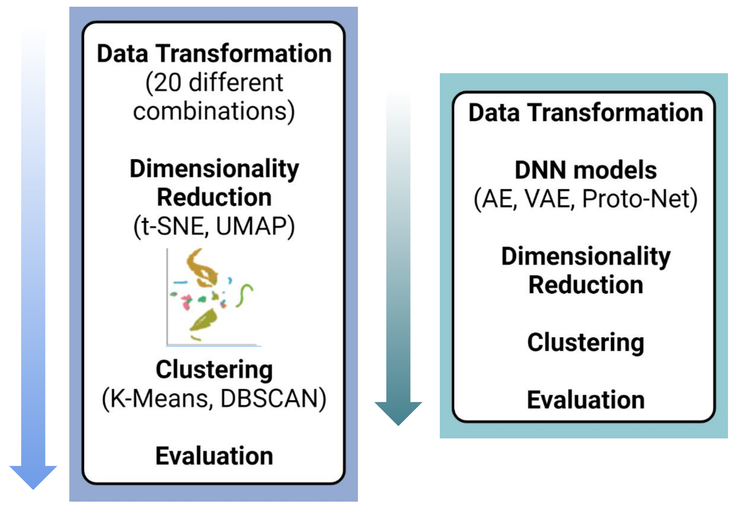
Data transformation methods used in recent studies

Tools	Preprocessing (data transformation used in the study or tool)
scVI	RAW
scLVM	RAW or log-linear fit
scGen	Total → Log
MNN	Deconvolution based normalization → Log
LIGER	Total → Meansquare
scImpute	Total → Log10
Scanorama	l2-norm
scIGANs	Minmax
ComBat-seq	RAW
DESC	Total → Log → Z-score
scMerge	Log → Z-score
scDHA	(Log2) → Minmax
scVAE	RAW
scGNN	Log
ICAnet	Total → Log2
scETM	RAW
iMAP	Log
scBatch	Log, (dataset with ERCC: scPLS) or (Raw → ComBat-seq)
Seurat V2, V3	Total(10000) → Log → Z-score
Harmony	Total(10000) → Log → Z-score
MARS	Total(10000) → Log → min(Z-score, 10)
Luecken2022	(scran) → Log

Study Overview

Human Pancreas Datasets: (GSE84133, GSE85241, E-MTAB-5061, GSE81608, GSE83139).

Mouse Cell Dataset: Tabula Muris



This study aims to investigate the effects of the various data transformation on public datasets and evaluate them with the most commonly used dimensionality reduction and clustering analysis.

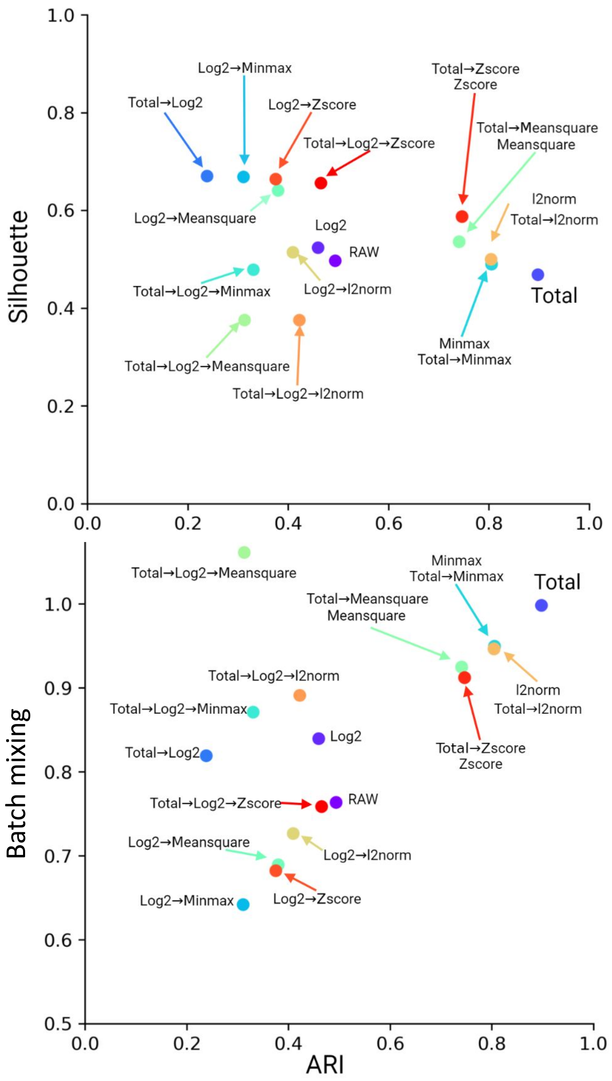
Results

Impact of data transformation on the multiple human pancreas datasets integrative task.

We aggregated all five datasets of the human pancreas and preprocessed them with 20 different combinations of data transformation.

Dataset	Number of class	Protocol	Accession ID
Baron (Human)	14	inDrop	GSE84133
Muraro	10	CEL-Seq2	GSE85241
Segerstolpe	14	SMART-Seq2	E-MTAB-5061
Xin	8	Fluidigm C1	GSE81608
Wang	7	SMART-seq	GSE83139

Here is the detailed investigation of the UMAP+DBSCAN. The total (transcripts per 20,000 transcripts) normalization method was the best performing preprocessing for Human Pancreas dataset.



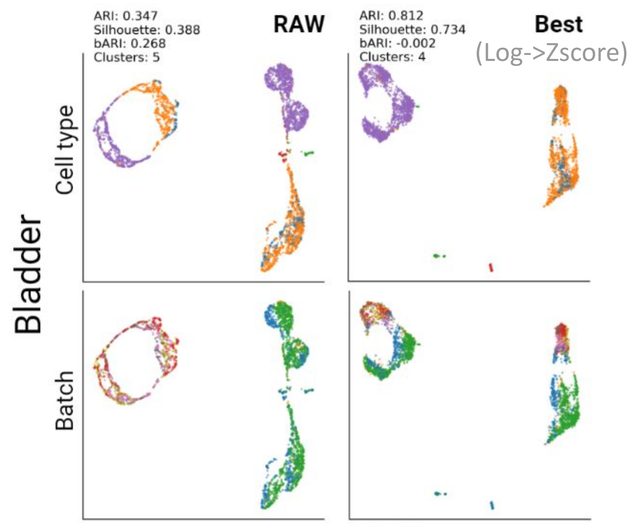
We compared the performance of our result to state-of-the-art methods benchmarked in Zhao2021. Notably, our result outperformed some of the tools benchmarked with ARI scores of 0.908 and 0.898.

Tools	MP	HP
Harmony ∗	0.969	0.955
Scanorama ∗	0.915	0.859
Seruat ∗	0.944	0.968
scVAE-GM ∗	0.805	NA
scVI ∗	0.932	0.759
LIGER ∗	0.914	0.911
scVI-LD ∗	0.875	0.656
scETM ∗	0.946	0.943
scETM $-\lambda$ ∗	0.851	0.474
scETM + adv ∗	0.944	0.946
Minmax + t-SNE+ DBSCAN	0.929	0.908
Total + UMAP+ DBSCAN	0.848	0.898
Meansquare + UMAP+ DBSCAN	0.903	0.741
$l2$-norm + UMAP+ DBSCAN	0.902	0.804
Total + AE + UMAP+ DBSCAN○	NA	0.947
Total + VAE + UMAP+ DBSCAN ○	NA	0.943

MP: Baron (Mouse) data
HP: Human pancreas datasets
∗ data are directly derived from the result by Zhao *et al.* (2021)
○ is trained with Baron, and picked best result of ten repeats

Impact of data transformation on the Tabula Muris datasets integrative task.

The Tabula Muris dataset consists of two different scRNA-seq protocols, SMART-Seq2 and CellRanger. By applying different data preprocessing, we observed significant improvements in 9 out of 11 tissue pairs of TM dataset. The cell types and batch ids are represented with different colors on each plot.



Impact of data transformation on the DNNs model.

The autoencoder (latent vector: 128) result without data transformation was ARI 0.551, but the 'Total' showed ARI 0.947. We investigated different DNNs models, Autoencoder, Variational Autoencoder, and ProtoTypical Network, and observed the similar impacts of preprocessing.

Next, we tested the power of DNNs models as a feature extractor for effective dimensionality reduction. We tested five different sizes for latent vectors from 2 to 32, and we could confirm that latent vector size between 8~32 is a reasonable length, and 50~100 is enough for latent vectors as reported in the other tools: scVI, scETM, scVAE, and scDHA.

	RAW	Total
AE-tSNE-KMean	0.315∼0.470	0.385∼0.588
AE-tSNE-DBSCAN	0.454∼0.521	0.468∼0.906
AE-UMAP-KMean	0.416∼0.521	0.550∼0.710
AE-UMAP-DBSCAN	0.299∼0.547	0.546∼0.870
VAE-tSNE-KMean	0.219∼0.346	0.432∼0.522
VAE-tSNE-DBSCAN	0.194∼0.618	0.558∼0.897
VAE-UMAP-KMean	0.229∼0.370	0.611∼0.785
VAE-UMAP-DBSCAN	0.063∼0.387	0.444∼0.894

The encoder/decoder has one hidden layer sized 1024. The size of latent layer is set to 128. Other parameters are tuned based on GridSearchCV with Skorch and scikit-learn. The range obtained with 10+ repeats of train and test.

Discussion

We demonstrated that, without any complex model for feature extraction or feature selection, optimized data transformation and dimensionality reduction methods show good performance in cell type classification. Moreover, our result shows that t-SNE and UMAP are still powerful tools for single-cell analysis.

The DNN models were able to compress gene expression profiles into very small sized vectors for further analysis.

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